

A Reinforced Particle Swarm Optimization Algorithm based on Beetle Antenna Search for UAV Path Planning

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Abstract: Traditional UAV path planning algorithms are difficult to achieve UAV path planning in complex 3-dimensional environments. In this paper, a UAV 3-dimensional path planning algorithm combining particle swarm optimization algorithm and beetle antenna search algorithm is proposed. Tent map inverse learning is introduced to initialize the algorithm, and B-spline curve is used to smooth the final formed paths. Experimental results show that the improved algorithm can obtain better paths faster in complex 3-dimensional environments.

Key Words: Path planning, Particle Swarm Optimization, Beetle Antennae Search, B-spline curve

1 Introduction

With the rapid development of robotics research, mobile robots such as unmanned vehicles, unmanned aerial vehicles (UAVs), and underwater robots have been widely used in various fields [1]. Mobile robots have been able to perform a variety of tasks in place of humans under certain conditions. UAVs have the advantages of small size, high mobility, and low cost, which have led to their application in many fields [2]. Path planning is an important part of UAV research [3]. UAV autonomous path planning is a complex optimization problem that involves multiple constraints, including performance and safety considerations. The ultimate goal is to generate an optimal mission route that connects the origin and destination while satisfying all the constraints[4]. Depending on the specific mission requirements, UAV path planning can be categorized into global path planning and real-time path planning. Global path planning leverages prior environmental knowledge to obtain an approximate optimal solution within a reasonable timeframe, while real-time path planning is a local optimization process that responds to unexpected threats and aims to find the optimal path within the shortest possible time.

Currently, methods for UAV path planning mainly include traditional control methods and intelligent control methods[5]. The traditional methods include the Artificial Potential Field Method, A* algorithm, and RRT method. Han[6] proposed an improved Artificial Potential Field Method, which achieves good obstacle avoidance by adjusting the angle and speed adjustment factors, but is not applicable to situations with many obstacles. Xiong[7] proposed a path planning algorithm combining A* algorithm and Dynamic Window Approach, which reduces the 3-dimensional environment obstacles to 2-dimensional obstacles and selects the key points of the global path as the sub-goals of local path planning. This method effectively improves the efficiency and path length of UAV path planning, but does not consider the flight altitude of the UAV. Gao[8] proposed a RRT method combining mixed population, which incor-

porates the population idea into the RRT method and effectively solves the problem of the RRT method easily falling into a local optimum, but the method has not been validated in a 3-dimensional environment.

Path planning for UAVs in complex environments is a challenging NP-hard problem. Traditional optimization algorithms have obvious advantages in small-scale spaces, but their computational complexity increases exponentially with the increase of space scale. Intelligent algorithms are widely used in UAV autonomous path planning problems in complex environments due to their high parallelism, strong robustness, and self-organization and self-learning characteristics. Currently, the commonly used intelligent algorithms for UAV path planning mainly include Genetic Algorithm, Ant Colony Optimization algorithm, and Particle Swarm Optimization algorithm. Huang[9] proposed an improved Genetic Algorithm that can plan shorter and smoother paths by improving selection operators, crossover operators, and variation operators. Experiments show that the method can plan a relatively smooth flight path in a 3-dimensional environment. However, the effectiveness of the method is somewhat limited when there are more obstacles in the environment. Additionally, Wei[10] proposed an improved Ant Colony Optimization algorithm for UAV 3-dimensional path planning. The method improves the local search strategy, initial pheromone adjustment factor, and adds a path offset factor to the heuristic function, thus reducing the complexity of the trajectory search space and improving the search efficiency and convergence speed of the algorithm. However, the path computed by the method is relatively tortuous.

Particle Swarm Optimization (PSO) algorithm has been widely applied to many fields due to its simple structure and fast running speed, which also makes it more applicable to path planning problems[11]. It can achieve fast path planning in relatively complex environments. However, when PSO algorithm is used to optimize high-dimensional functions, it has some disadvantages such as low optimization accuracy, insufficient global search ability, easy to fall into local optimal solutions, or need to introduce more particles[12]. These drawbacks also make the path planning results computed by PSO often not optimal paths. In order

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to solve these problems, many scholars have improved the particle swarm algorithm in recent years. Zou[13] proposed the clustering and foraging behavior of artificial fish swarm algorithm based on PSO, which makes the new hybrid algorithm have better global search ability. Wu[14] proposed to introduce the Metropolis criterion in simulated annealing into PSO algorithm, making it easier to jump out of local optimal solutions, and combined with RTS method to obtain better paths. However, it also increases the computation time of path planning.

The Beetle Antenna Search (BAS) algorithm is a single machine optimization algorithm proposed by Jiang in 2017 to find the optimal value by simulating the foraging behavior of beetles in nature[15]. Due to its simple structure, it has a fast computational speed. In recent years, scholars have applied this algorithm to path planning. Zhao[16] proposed an algorithm combining A* algorithm and BAS algorithm. The path is generated with the A* algorithm and then optimized with the BAS algorithm. Experiments show that the algorithm can obtain better paths. Wu[17] proposed a path planning algorithm combining the BAS algorithm and the APF algorithm. Experiments show that this algorithm can generate a feasible path quickly, but the paths generated by the algorithm are usually long and tortuous. In general, the BAS algorithm has powerful local search capability and fast convergence speed. However, it lacks global search capability.

To quickly calculate a short and smooth flight path of UAVs in complex 3-dimensional environments, this paper proposes a PPSOBAS algorithm that combines the PSO algorithm and BAS algorithm. The PSO algorithm has a strong global optimization capability, while the BAS algorithm has a strong local optimization capability and fast computational speed when solving path planning problems. First, initial paths are generated using the Tent map. Next, individuals of the BAS algorithm are used to replace the individuals in the PSO algorithm. Global optimization is achieved using the population search effect of the PSO algorithm. The variation of particles is implemented by the BAS algorithm, where each generation of particles uses BAS to explore their surrounding space before performing global optimization search and retaining the better of the particles themselves and the mutated one. This enhances the global exploration capability and local optimization capability of the PSO algorithm to further optimize the generated paths. Finally, the calculated paths are smoothed by the B-spline curve to make them smoother and more feasible. Since both the PSO algorithm and BAS algorithm have fast computational speed, the new algorithm can also solve the problem quickly. Experiments in Part 4 prove that the PPSOBAS algorithm can obtain better paths quickly when calculating path planning problems in complex 3-dimensional environments.

2 Model for UAVs path planning problem

2.1 Environmental Modeling

The path planning problem for UAVs is to find a feasible path from the starting point to the end point in a 3-dimensional environment, while considering various constraints such as obstacles, flight altitude, flight time, and energy consumption. UAVs generally work within a specific

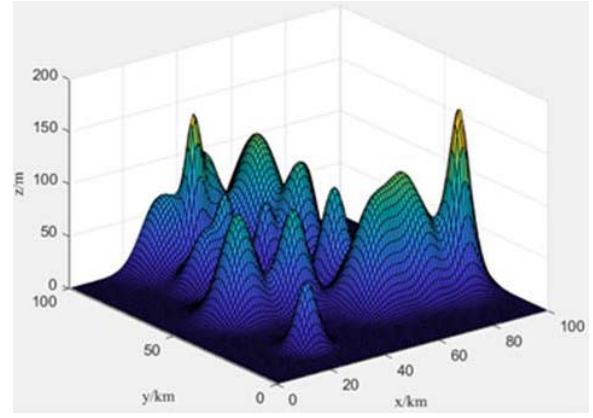


Fig. 1: environmental modeling

range and cannot fly outside the boundary of the environment. The mission space can be described as:

$$\Omega = \{(x, y, z) \mid 0 \leq x \leq x_{\max}, 0 \leq y \leq y_{\max}, 0 \leq z \leq z_{\max}\} \quad (1)$$

Obstacles and other threats in the mission space can be equated to peaks, after fusion with the digital map. The mathematical model of the mountain peaks is as follows:

$$z(x, y) = \sum_{i=1}^n h(i) \times \exp \left[-\frac{(x - x_0(i))^2}{x_i(i)} - \frac{(y - y_0(i))^2}{y_i(i)} \right] \quad (2)$$

where $x_0(i)$, $y_0(i)$ are the coordinates of the center of the i th mountain, $x_i(i)$, $y_i(i)$ are the decrements of the i th mountain along the x and y axes, $h(i)$ is the height of the i th mountain, and n is the number of mountains. The actual working environment often has a great number of threats and obstacles. Increased hazards and obstacles can create more disruptions in the path planning process, which will also result in the algorithm requiring more computation time. In order to more realistically simulate the working environment of the UAV and verify the path planning effect of the algorithm in a complex environment, 20 obstacles were randomly generated in the mission space, differing from the 7 obstacles in [14], and the simulated map is shown in Fig. 1.

2.2 Constraints

Path planning is the process of finding a safe and feasible route from the start point to the end point that can reach the end point with the shortest distance and avoiding terrain obstacles while satisfying the physical constraints of the vehicle itself. The environmental constraints and the physical constraints of the vehicle constitute the two main types of constraints in the path planning process.

The flight path should be within the mission space to ensure that the vehicle does not fly away from the mission space. Besides, the flight altitude of the vehicle should always be above the obstacle altitude due to the presence of obstacles in the mission space:

$$\begin{cases} x_{\min} \leq x_i \leq x_{\max} \\ y_{\min} \leq y_i \leq y_{\max} \\ z_{\min} \leq z_i \leq z_{\max} \\ z_i \geq z_m(x_i, y_i) \end{cases} \quad i = 1, 2, \dots, n \quad (3)$$

where x_i, y_i, z_i represent the coordinates on the flight path, z_m represents the height of the obstacle corresponding to the horizontal and vertical.

The physical capabilities of the vehicle are an important consideration in the path planning process. These physical constraints are mainly reflected in Table 1, where $a_i = (x_i - x_{i-1}, y_i - y_{i-1})^T$, $b_i = (z_i - z_{i-1})$.

Table 1: physical constraints

Num	Constraints	Equations
1	Pitch angle	$\alpha = \arccos \frac{a_i^T a_{i+1}}{ a_i a_{i+1} }$
2	Yaw angle	$\tan \theta = \frac{ b_i }{ a_i }$
3	track length	$l_{min} < l_i < l_{max}$

2.3 Fitness Function Modeling

The length of the path and the flight altitude of the vehicle are the most important factors affecting the flight of the vehicle. The shortest path gets to the destination faster and consumes less flight time and fuel, which allows the vehicle to perform more missions. The lowest altitude increases the probability that the vehicle will not be detected, making the flight safer. Therefore, we consider both path length and flight altitude to form a fitness function according to a linear combination:

$$F = \sum_{i=1}^m (\lambda_1 L_i + \lambda_2 H_i) \quad (4)$$

where L_i represents the distance between the i th neighboring two points, H_i represents the flight altitude of the i th point, λ_1 and λ_2 is the coefficient of the length and altitude, which are constant values. The linear combination of length and height allows for more intuitive differentiation of paths. The smaller value of $F()$ means that the path is shorter and safer.

3 Improving Particle Swarm Algorithm

3.1 Tent map backward learning initialized populations

In the particle swarm algorithm, the convergence performance of the algorithm is affected by the initial population[18]. The larger and more uniformly distributed the population is, the more the algorithm can converge to the optimal solution in a shorter time; on the contrary, it will affect the convergence performance of the algorithm. The initial population using chaotic map is random, ergodic and bounded, which can effectively improve the search efficiency of the algorithm. The Tent map produces a more uniform initial sequence than the Logistic map, so this paper uses the Tent map to initialize the particle population, and uses the backward learning strategy to optimize the initial population. By competing with the existing individuals, the better individuals are selected for the next generation of learning, which can expand the search range of the population and reduce the invalid search, thus improving the convergence speed of the algorithm. The mathematical expression of the Tent map is:

$$x_{n+1} = \begin{cases} \frac{x_n}{a}, & 0 < x_n \leq a \\ \frac{1-x_n}{1-a}, & a < x_n \leq 1 \end{cases} \quad (5)$$

where a is the chaotic coefficient in the range of $[0, 1]$. The inverse solution is defined as a feasible solution in d -dimensional space as $x = (x_1, x_2, \dots, x_d)$, $x = [a, b]$. The inverse solution is $x' = (x'_1, x'_2, \dots, x'_d)$, where $x'_i = a + b - x_i$.

The specific step of initializing the population using Tent map backward learning is to generate the positions of M particle populations in the search space using Tent map as the initial population A $X_{ij}, i = 1, 2, 3 \dots d, j = 1, 2, 3 \dots M$. Based on the definition of the inverse solution, the inverse population of each aspen population in the initial population A was generated as the inverse population B. The populations A and B were combined, and the $2M$ aspen populations were ranked in ascending order using the fitness values, and the aspen population with the top M fitness values was selected as the initial population.

3.2 Particle Sarm Optimization

Particle swarm optimization (PSO) is a population intelligence affine algorithm proposed by Kennedy and Eberhart in 1995, which can effectively solve some typical function optimization problems. Compared with traditional optimization algorithms, PSO has faster computational speed and better global search capability. Therefore, it has certain advantages in solving problems such as path planning. Suppose there are M particles, each particle represents a path, each path contains k points, and the inter-particle update is:

$$v_i^{k+1} = w v_i^k + c_1 r_1 (p_{ibest}^k - x_i^k) + c_2 r_2 (g_{ibest}^k - x_i^k) \quad (6)$$

$$x_i^{k+1} = x_i^k + v_i^{k+1} \quad (7)$$

where $i = 1, 2, 3 \dots M, k = 1, 2, 3 \dots N$ denotes the current number of iterations. w is the inertia factor, which is a non-negative number and plays a great role in the convergence of the algorithm. the larger its value, the wider the range of particle leaps, the easier it is to find the global optimum, but it also misses the ability of local search. p_{ibest} , g_{ibest} are the local and global optimum, respectively position. The acceleration constants c_1 and c_2 are non-negative constants that adjust the weights of the local optimum and the global optimum. r_1 and r_2 are random numbers in the range of $[0, 1]$.

Besides, PSO with linearly decreased inertia weight w , cognitive and social coefficients c_1 and c_2 on iteration generations have been introduced. Therefore, the various weighting factors can be calculated as follows:

$$\begin{aligned} w &= w_{max} - \frac{w_{max} - w_{min}}{N} \times k \\ c_1 &= c_{1max} - \frac{c_{1max} - c_{1min}}{N} \times k \\ c_2 &= c_{2max} - \frac{c_{2max} - c_{2min}}{N} \times k \end{aligned} \quad (8)$$

where $w_{max}, c_{1max}, c_{2max}$ and $w_{min}, c_{1min}, c_{2min}$ are the maximum and minimum values of inertia weight, cognitive coefficient and social coefficient respectively. If the inertia weight, cognitive coefficient and social coefficient are larger, PSO will have a stronger global exploration ability. On the other hand, if the coefficient is smaller, the local exploitation ability will be stronger.

3.3 Beetle Antennae Search

Beetle Antennae Search (BAS) algorithm is an intelligent optimization algorithm proposed by Jiang in 2017, which

simulates the search of an aspen when seeking food, and is a monadic search algorithm with simple principles, few parameters, and low computational effort, which has advantages in dealing with low-dimensional optimization objectives. The position of the beetle in the k -dimensional space is:

$$X(x_1, x_2, x_3, \dots, x_k) \quad (9)$$

The position of the beetle's left and right tentacles:

$$\begin{cases} X_r = X + l * \bar{d} \\ X_l = X - l * \bar{d} \end{cases} \quad (10)$$

where l denotes the distance between the center of mass of the aspen and the tentacles, $\bar{d}$ denotes the normalized random unit vector:

$$\bar{d} = \frac{\text{rands}(D, 1)}{\|\text{rands}(D, 1)\|} \quad (11)$$

After the two tentacles obtain information about the objective function of the two positions, then compare the fitness of the two positions and choose the direction toward the better position:

$$X_{t+1} = X_t + \delta_t * \bar{d} * \text{sign}[f(X_r) - f(X_l)] \quad (12)$$

$$\delta_{t+1} = \alpha * \delta_t \quad (13)$$

where t denotes the number of iterations, $f()$ is the fitness function and $\text{sign}()$ is the sign function, δ_t denotes the step size at the t th exploration, usually the step size needs to become smaller during each iteration.

3.4 B-Spline Curve

Each particle in the algorithm represents a collection of multiple points. Assume that each particle contains $n + 1$ points $P_0, P_1, \dots, P_n$ and the generated B-spline curve is a k th curve. The list $t_i (i = 1, 2, 3 \dots n + k + 1)$ containing $n + k + 1$ nodes is constructed in the space, and the k th B-spline curve obtained is:

$$P(t) = \sum_{i=0}^n P_i F_{i,k}(t) \quad (14)$$

where P_i is the curve equation corresponding to the i th control point, $F_{i,k}(t)$ is the k th order B-spline curve basis function. It can be solved by the Cox-deBoor recursive formula:

$$F_{i,1}(t) = \begin{cases} 1, & t_i \leq t \leq t_{i+1} \\ 0, & \text{others} \end{cases} \quad (15)$$

$$F_{i,k}(t) = \frac{t - t_i}{t_{k+i} - t_i} F_{i,k-1}(t) + \frac{t_{k+i+1} - t}{t_{k+i+1} - t_{i+1}} F_{i+1,k-1}(t) \quad (16)$$

When $t_{i+1} - t_i$ is a constant value, it is a uniform B-spline curve. Suppose $Q_i, Q_{i+1}, Q_{i+2}, Q_{i+3}$ are four adjacent nodes and the B-spline curve of the i th segment is:

$$P_i(x) = \frac{1}{6} \begin{bmatrix} 1 \\ x \\ x^2 \\ x^3 \end{bmatrix}^T \begin{bmatrix} 1 & 4 & 1 & 0 \\ -3 & 0 & 3 & 0 \\ 3 & -6 & 3 & 0 \\ -1 & 3 & -3 & 1 \end{bmatrix} \begin{bmatrix} Q_i \\ Q_{i+1} \\ Q_{i+2} \\ Q_{i+3} \end{bmatrix} \quad (17)$$

Assuming that the coordinates of the control points in the flight path are denoted by (X_i, Y_i, Z_i) ($i = 1, 2, \dots, n$),

where the parameter x is uniformly taken as all numbers between 0 and 1 with an interval of 0.01, the i th section of the B-spline curve is obtained. The coordinates of any point of the curve (X_t, Y_t, Z_t) are expressed as follows:

$$\begin{cases} X_t = \sum_{i=0}^m X_i F_{i,k}(t) \\ Y_t = \sum_{i=0}^m Y_i F_{i,k}(t) \\ Z_t = \sum_{i=0}^m Z_i F_{i,k}(t) \end{cases} \quad (18)$$

3.5 Combination of PSO, BAS, Tent map backwards learning and B-spline curve

The PSO algorithm has a strong global optimization capability, while the BAS algorithm has a strong local optimization capability and fast computational speed when solving path planning problems. Considering the advantages of the PSO algorithm and the BAS algorithm, this paper proposes a PPSOBAS algorithm for UAV path planning in a complex dimensional environment. The algorithm proposed in [19] combines BAS and PSO, and incorporates the direction of insect olfaction as part of the particle swarm velocity update. While this algorithm increases the judgment of the particle swarm towards the surrounding environment, it does not allow the PSO to escape from local optima effectively. In contrast, the PPSOBAS algorithm proposed in this paper combines the Tent map, PSO, BAS, and B-spline curve. Firstly, the Tent map backward learning is used to initialize the particle population, making the particles more evenly distributed throughout the entire task space. Secondly, the BAS individual replaces the particle in PSO, and each individual performs a BAS operation before the group optimization. The particle with better fitness after the operation replaces the original particle for group optimization. Each generation of particles performs a mutation operation by the BAS algorithm, which can effectively escape local optima. Additionally, an adaptive step length adjustment is also used to enhance the convergence accuracy of PPSOBAS. Finally, the generated path is fitted by a B-spline curve to make the calculated path smoother. The algorithm flow is shown in Algorithm 1.

Remark 1 Compared with the algorithm in [19], The algorithm in this paper will run the BAS algorithm for each generation of particles to explore their surrounding space before performing a population search. Particles with better fitness will replace the original particles. This variation increases the chance of getting rid of the local optimum and finding the optimal path, which allows to obtain better paths with fewer particles. The adjustment of the parameters w, c_1, c_2 makes the algorithm have higher convergence accuracy in the later stage, which allows the algorithm to obtain shorter and safer paths. Besides, The reduction of particles will also reduce the computation time of the algorithm.

4 Algorithm simulation and result analysis

This experiment is simulated on matlab2021b, and the CPU used in the computer is intel i7-10700. In this experiment, the number of iterations N is 100 and the number of particles M is 50. The initial parameters of the PSO are: $w_{max} = 1, w_{min} = 0.4, c_{1max} = 1.2, c_{2max} = 1.4, c_{1min} = 0.6, c_{2min} = 0.6$. The initial parameters of the PSO are: $\alpha = 0.95, \delta_t = 1$. To verify the effectiveness of the algorithm, the PPSOBA algorithm mentioned in

Algorithm 1 PPSOBAS design process**Input:** mode of environment and constrains.**Output:** a feasible solution for path planning problem.

- 1 Initialize M particles using equation (5), each particles is composed of a swarm of d -dimension positions of particles $x_{i,k}$ and d -dimension speed vectors $v_{i,k}$;
- 2 Using the B-spline curve to fit the points in $x_{i,k}$ into a path;
- 3 Using equation (4) Calculate, compare and store the p_{ibest} and g_{ibest} ;
- 4 **while** Less than the number of iterations **do**
 - 5 Use equation (12) to update the position for all particles;
 - 6 Evaluate the fitness of the updated particles to the original particles;
 - 7 Fitting of new particles using the B-spline curve;
 - 8 The updated particle with less fitness replaces the original particle position;
 - 9 Evolve $v_{i,k+1}$ ($i = l, \dots, m, k = l, \dots, n$) with the equation (7);
 - 10 Get $x_{i,k+1}$ with the equation (6);
 - 11 Evaluate x_{k+1} using the equation (4);
 - 12 Update the position of pbest and gbest;
 - 13 $i=i+1$;
 - 14 Update the inertia weight, cognitive coefficients and social coefficients using the equation (12);
 - 15 Update the δ_t using the equation(13);
- 16 Return the best solution of the swarm.

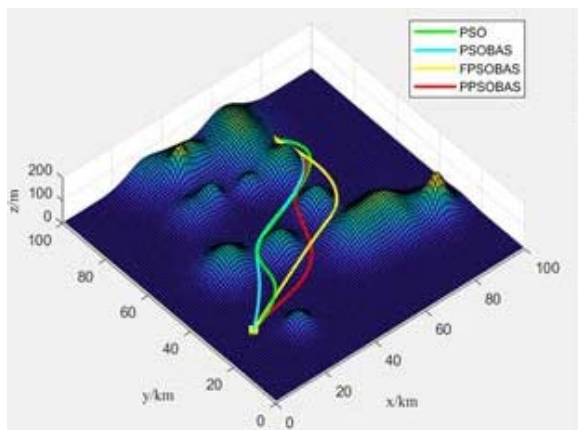


Fig. 2: path planning for path1 by different algorithms

this paper is compared with the traditional PSO algorithm, the PSOBAS algorithm without parameter adjustment, and the FPSOBAS algorithm from [19]. In order to verify its path planning effect in a complex environment, this experiment is designed for three different cases of path planning, and Fig. 2 shows the comparative graph of path planning results for path1. The four color lines represent the final paths calculated by the four algorithms. In order to observe the advantages and disadvantages of the four algorithms more intuitively, Fig. 3 shows the comparison of the fitness functions calculated for path 1. All algorithms complete the path planning. However, by comparing the images we can find that the fitness function values obtained by PPSOBAS algorithm are much smaller than the other three algorithms, which indicates that the PPSOBAS algorithm can achieve better results in the path planning problem, and the paths obtained are shorter and more secure.

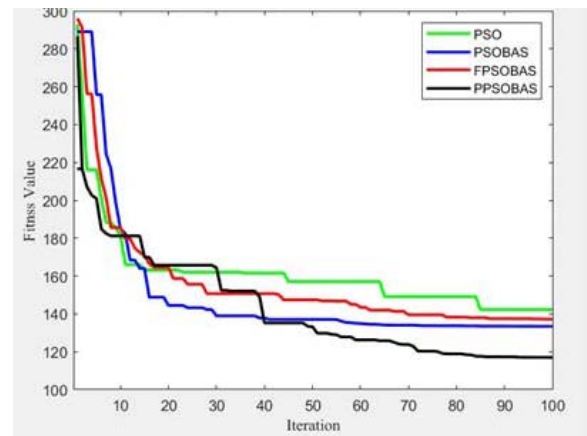


Fig. 3: the fitness values change with the iteration

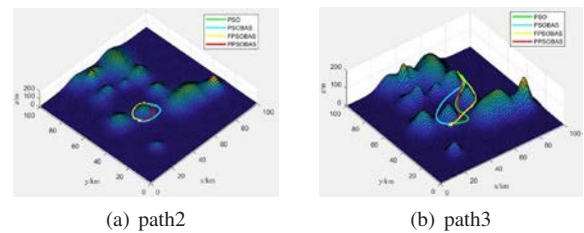


Fig. 4: experimental results with different paths

To demonstrate the effectiveness of the PPSOBAS algorithm in path planning for different distances, the starting and target points of the paths were changed on the map. Fig. 4 shows the path planning and adaptation functions for single-peak and multi-peaks. By comparison we can find that the PPSOBAS algorithm achieves better results for both short distances and multiple peaks. Table 2 shows the comparison of the final fitness values of different paths.

Table 2: The fitness value in different paths

Path	PSO	PSOBAS	FPSOBAS	PPSOBAS
path1	154.8089	133.4667	137.2592	115.0776
path2	115.7534	105.7114	100.4373	93.9467
path3	74.2292	66.5291	66.19	61.5146

In the above experiments, we demonstrate that PPSOBAS can effectively avoid local optima and achieve better results when solving path planning problems. Since the global search capability of the algorithm is further enhanced by replacing a single beetle with a single particle, the same optimization effect can be achieved with fewer particles when solving path planning problems. The reduction in the number of particles also significantly reduces the time required for operations, allowing for faster computational speed. To verify that the method is both good at path planning and fast, we reduced the number of particles to 30 and then computed all four algorithms for path1 50 times, taking the average of the fitness values and computation times, and Table 3 shows their comparison.

The comparison in Table 3 shows that when the number of particles of the PPSOBAS algorithm is reduced, the path planning results do not change very much, but with the reduction of the number of computations, the compu-

Table 3: The e average of the fitness values and computation times

Algorithm	PSO	PSOBAS	FPSOBAS	PPSOBAS
fitness	151.3041	135.4258	136.5932	116.7116
time	84.4467	101.8991	118.7092	79.8606

tation time decreases significantly compared with the other two methods, which proves that the PPSOBAS method can achieve better path planning results with fewer particles and faster computation speed. It can be better applied in solving practical problems.

5 Conclusions

In this paper, a path planning algorithm that incorporates the PSO algorithm, BAS algorithm, Tent map and B-spline curve is proposed. The PSO algorithm is enhanced by the BAS algorithm, which enables fast path planning for UAVs in complex 3-dimensional environments. The experiments in the part 4 verify the effectiveness of the algorithm by comparing it with other algorithms. Experiments show that it can compute better paths for different start and end points with less computation time than other algorithms in complex 3-dimensional environments.

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